

The empirical recurrence for OEIS A238651: recovering a conjecture that is not written down

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Abstract

OEIS A238651 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 272$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A238651 is “Number of $(n+2) \times (5+2)$ 0..1 arrays with no element equal to all horizontal neighbors or equal to all vertical neighbors, and new values 0..1 introduced in row major order.”. It has offset 1 and begins

8, 128, 630, 4742, 30116, 204216, 1349732, 9010526, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:53:26, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 272$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 272$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 544$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	5	c_{18}	−118649209	c_{35}	−236900824	c_{52}	18292726
c_2	30	c_{19}	−129233247	c_{36}	1635617318	c_{53}	6052559
c_3	−37	c_{20}	265480066	c_{37}	265536328	c_{54}	−5754977
c_4	−632	c_{21}	321333931	c_{38}	−1073688981	c_{55}	−2174114
c_5	−1082	c_{22}	−534927504	c_{39}	−347634219	c_{56}	1403870
c_6	6391	c_{23}	−640842205	c_{40}	720489296	c_{57}	645516
c_7	23663	c_{24}	996096271	c_{41}	348641390	c_{58}	−246818
c_8	−13767	c_{25}	1023274144	c_{42}	−502789927	c_{59}	−147117
c_9	−215778	c_{26}	−1671606678	c_{43}	−266097832	c_{60}	27298
c_{10}	−259880	c_{27}	−1273545411	c_{44}	337392810	c_{61}	24309
c_{11}	1012178	c_{28}	2408287384	c_{45}	162155156	c_{62}	−1039
c_{12}	2685731	c_{29}	1199700086	c_{46}	−202037982	c_{63}	−2722
c_{13}	−1852047	c_{30}	−2883021032	c_{47}	−82283289	c_{64}	−177
c_{14}	−13414398	c_{31}	−831247697	c_{48}	105308039	c_{65}	179
c_{15}	−4442528	c_{32}	2838564699	c_{49}	36601879	c_{66}	26
c_{16}	45173212	c_{33}	429235162	c_{50}	−47575605	c_{67}	−5
c_{17}	37601107	c_{34}	−2320839767	c_{51}	−15220362	c_{68}	−1

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A238651.
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